Analyzing Economic Indicators and Global Health Outcomes

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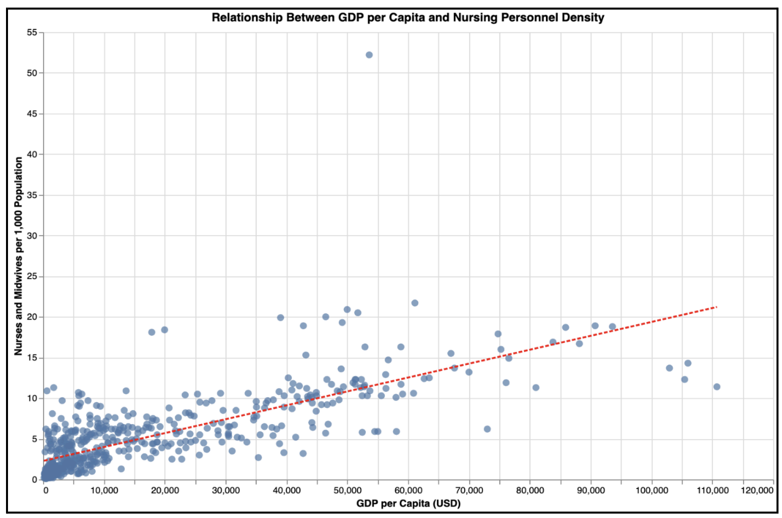
Introduction

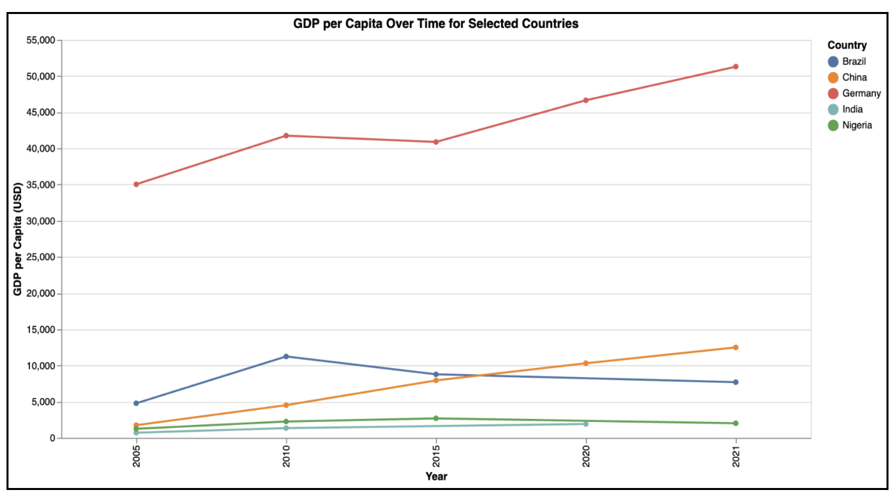
The team's motivation grew from mounting international awareness about health inequality together with economic influences that determine public health results. This research aims to analyze the relationship between how different levels of national wealth impacts essential health indicators. Our goal utilized data science methods to transform economic relationships into quantifiable forms in order to discover patterns about health across different nations. Our project explored the connection between national economical conditions, healthcare staffing levels, and patient health outcomes. Our analysis considers the regional variations of economic status together with healthcare personnel availability and health outcomes distributions. The economic data is from the United Nations Statistics Division [2] [3] and the health data is from the World Health Organization Global Health Observatory [1].

Insights and Visualizations

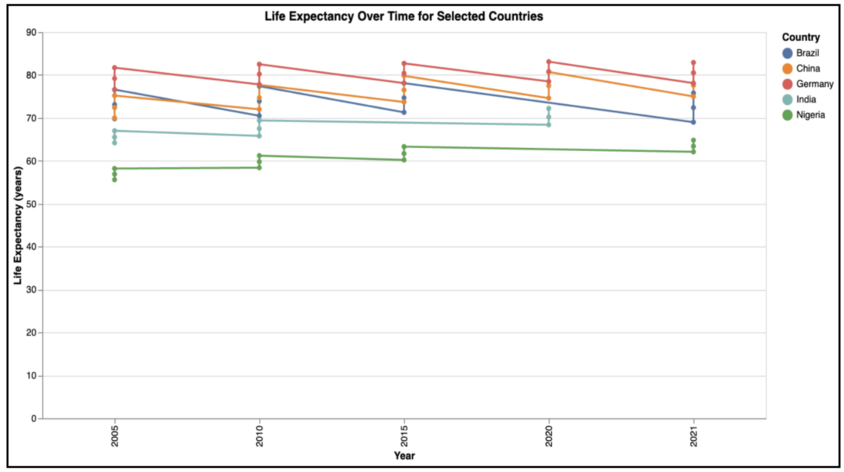
Our initial scatter plot revealed a positive correlation between GDP per capita and nursing personnel density. Countries with higher GDP consistently relate to larger healthcare workforces. The regression line indicated that economic growth is a strong predictor of healthcare staffing levels, though variability exists most likely due to regional and political differences. This connection shows that wealth boosts funding for health infrastructure that remains essential for enhancing public health results.



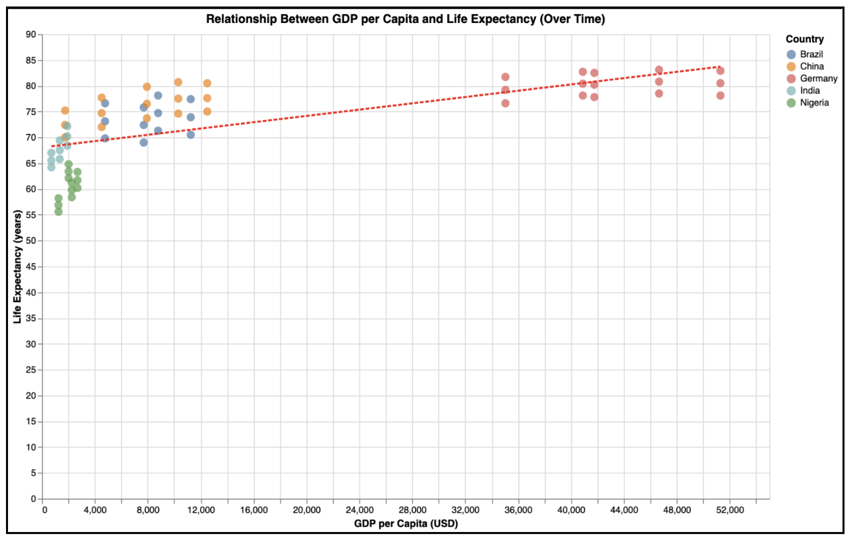
Our GDP per capita line graph highlights global disparities in economic growth. These five countries displayed were chosen to provide a diverse representation of global economies and health systems across different continents and development stages. Using this slimmed down subset of countries helped us cut out the noise of displaying all countries in the dataset. Germany consistently leads with the highest GDP, reflecting stable economic conditions. China shows rapid growth, pointing to accelerated development. Brazil peaks early but declines, suggesting economic instability. India improves gradually, while Nigeria sees little change. These trends show there is unequal economic progress.



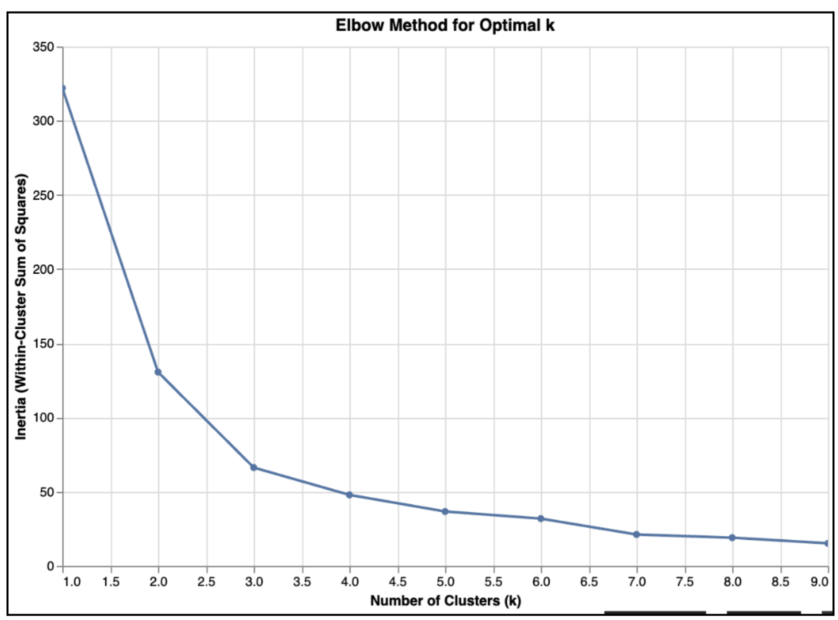
Our life expectancy line graph highlights disparities in health outcomes across countries over time. Germany consistently leads with the highest life expectancy, reflecting strong healthcare systems and living conditions. China and Brazil follow, though Brazil shows a decline in recent years. India sees steady improvement, while Nigeria remains the lowest with minimal change. The graph suggests that stronger economies are often linked to better health outcomes, while lower income countries continue to have persistent health challenges.



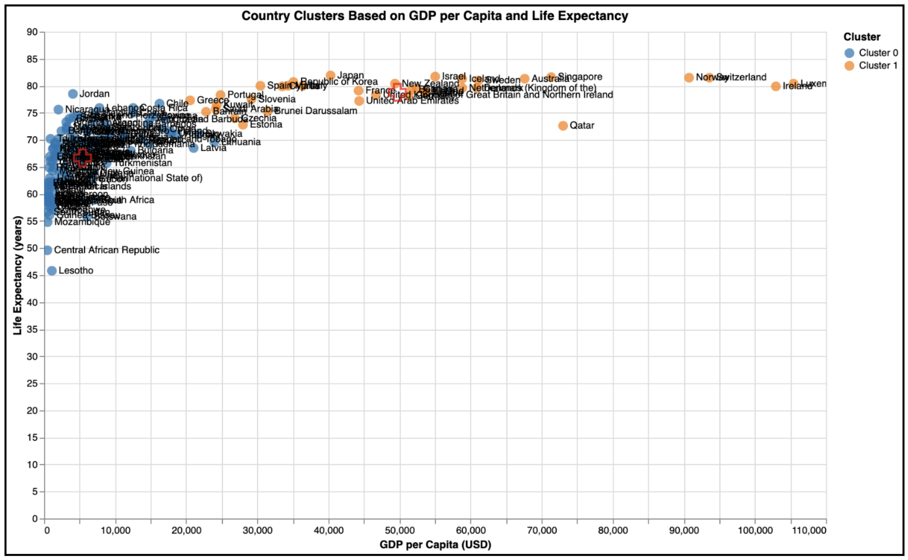
Our scatter plot examining GDP per capita and life expectancy over time reveals a clear positive correlation between the two variables. Countries with higher GDP per capita, such as Germany, consistently report longer life expectancies. Countries with lower GDPs like Nigeria and India tend to have shorter lifespans. The upward sloping regression line reinforces this relationship, suggesting that economic strength is a reliable predictor of public health outcomes. However, some variation exists, indicating additional factors such as healthcare infrastructure or policy also influence life expectancy beyond just income level.



Our elbow graph helps determine the optimal number of clusters for our k-means model. The sharp drop in inertia between k=1 and k=2 suggests that adding more clusters significantly improves model fit early on. The curve begins to level out after k=2, pointing towards diminishing returns. Based on this elbow point, k=2 appears to be the best number of clusters for grouping countries by their economic and health indicators, balancing the model accuracy and simplicity.



Our clustering visualization groups countries based on GDP per capita and life expectancy, revealing two distinct clusters. Cluster 1 has countries with high income and high life expectancy, showing strong health systems and economic stability. Cluster 0 contains countries with lower GDP and shorter life spans. This separation emphasizes the global divide in both economic and public health. The clustering supports our earlier findings, showing that countries with greater economic resources tend to provide better health outcomes for their populations. All of the countries in Cluster 1 have a GDP per capita of $20,000 USD or higher and had life expectancies of around 80 years with none being substantially higher than the others. From the graph, we can conclude there is a logarithmic trend, and the maximum threshold to a country's life expectancy of around 83 years that will not increase no matter how high its GDP per capita is. This is demonstrated as countries with a lower GDP per capita in Cluster 1 such as Japan and Slovenia have similar life expectancies to Ireland and Luxembourg, the two countries with the highest GDP per capita.



Conclusions and Takeaways

- Data driven monitoring and adaptive policymaking amplify impact.
    - Setting clear KPIs such as vaccination coverage and mortality rates for children under the age of 5 would help establish real time dashboards. Applying statistical inference such as confidence intervals and tend-break tests allows people to set aside their resources for the highest performing interventions. This also makes sure the current strategies in place remain effective as the conditions develop.
- Cross Sector strategies are imperative for economic growth
    - There are various sectors that contribute to a growing economy because aligning educational, health, sanitation and social services policies ensures that there is economic growth for sustainable gains in the world population.
- Investing early in preventive care.
    - Regression predicts that a marginal increase to the GDP per capita aligns with a huge improvement in life expectancy. Putting resources toward immunization and healthcare will directly result in health and economic stability
- Economic strength is a key driver of health outcomes.
    - Across all visualizations, higher GDP per capita consistently correlates with better life expectancy and larger healthcare workforces. Wealthier nations are generally able to invest more in public health infrastructure and services.
- Global health and economic disparities persist.
    - The gap between high income and low income countries remains significant. While countries like Germany and Singapore enjoy both economic and health advantages, nations such as Nigeria and Lesotho continue to face barriers in both domains.
- Economic growth does not guarantee consistent health improvements.
    - Some countries, such as Brazil, demonstrate that even middle-income or growing economies can experience setbacks in health outcomes, highlighting the influence of policy, stability, and healthcare access.

Policy Recommendation - Next Steps

- Establish integrated Health & Socio Economic Data Dashboards
  - Building a public dashboard that integrates GDP per capita, life expectancy, and other education metrics which are updated quarterly, will help the decision makers to allocate resources efficiently and measure intervention impact.
  - Led by national statistical offices, WHO's Health Metric and Evaluation and UN Global Pulse; this would be funded by the Global Partnership for sustainable development data. Approximately a 6 month pipeline is estimated for getting the first version live and usable with end to end testing and trained staff.
- Launch bi-annual health literacy & behavior change patterns in less developed countries
  - Having a health drive that focuses on nutrition, hygiene, childcare which are tailored to the lowest GDP-clusters identified by k model.
  - Being more educated will motivate people to take on new health practices and reduce disease burdens within the local community.
  - UNICEF regional officers would take lead and by partnering with the national Ministries of Education and Health, they can integrate simple curricula modules into the public school systems

References

[1] World Health Organization. 2025. *Life expectancy at birth (years)*. https://data.who.int/indicators/i/A21CFC2/90E2E48 (Accessed March 9, 2025)

[2] United Nations Statistics Division. (2023). GDP and GDP per capita. UNdata. Retrieved March 9, 2025, from https://data.un.org/_Docs/SYB/CSV/SYB67_230_202411_GDP%20and%20GDP%20Per%20Capita.csv

[3] United Nations Statistics Division. (2023). Health personnel. UNdata. Retrieved March 9, 2025, from https://data.un.org/_Docs/SYB/CSV/SYB67_154_202411_Health%20Personnel.csv